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T.E. (Computer Engineering)

WEB TECHNOLOGY

(2019 Pattern) (Semester - II) (310252)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right side indicate full marks.

Q1) a) Explain doGet() & doPost() methods of servlet. Differentiate doGet Vs doPost (Min 04). **[9]**

b) What is XML DTDs? Explain with example. Differentiate XML DTDs Vs XML schema (Min.04). **[9]**

OR

Q2) a) Explain the servlet lifecycle. Explain session management using cookies and URL Rewriting. **[12]**

b) Write note on AJAX. **[6]**

Q3) a) Explain the JSP support for MVC paradigm. **[8]**

b) Explain struts framework with respect to architecture, actions, interceptors & exception handling. **[9]**

OR

Q4) a) Explain JSP lifecycle. Differentiate JSP Vs Servlet. (Min.04) **[9]**

b) Explain the concept of web services. Explain in brief WSDL & SOAP. **[8]**

Q5) a) Explain various types of Arrays in PHP. Explain each with example code. **[9]**

b) Explain the following :

i) WAP & WML **[4]**

ii) C# Vs Java **[5]**

OR

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Q6) a) Explain the PHP with MySQL using example. **[6]**

b) Write note on : **[12]**

i) Session tracking in PHP.

ii) NET framework

iii) NodeJS

Q7) a) Explain Ruby with its advantages. Explain control statements in Ruby. **[10]**

b) Explain EJB concept & five basic example of using EJB. **[7]**

OR

Q8) a) Explain the arrays in Ruby. Explain Rails with AJAX. **[10]**

b) Explain Document Request in Rails. **[4]**

c) Explain advantages of Ruby and Rails. **[3]**

x

x

x

Solution:

Q1 (a) Explain doGet() & doPost() methods of servlet. Differentiate doGet vs doPost (Min 4). [9 Marks]

Ans:

doGet() and doPost() are two important methods of HttpServlet used to handle **HTTP GET** and **HTTP POST** requests from the client.

1. doGet() Method – Explanation

1. Handles HTTP GET Request

This method is called when data is sent through the URL or directly using a hyperlink.

2. Visible Parameters

All values appear in the URL as *query strings*.

Example:

example.com/page?name=vijay

3. Limited Data Sending

Because URLs have length limits (approx. 1024 chars).

4. Used for Reading/Retrieving Data

Best for non-sensitive data like search queries, filters, viewing pages.

5. Not Secure

Data can be bookmarked, cached, and visible in browser history.

2. doPost() Method – Explanation

1. Handles HTTP POST Request

Called when data is submitted through a form using method="post".

2. Data Not Visible in URL

Values are sent inside the body of the request → safer and private.

3. Can Send Large Amount of Data

Suitable for uploading files, login pages, registration forms, etc.

4. Secure Compared to GET

Data is not cached or stored in browser history.

5. Mostly Used for Updating/Submitting Data

Used for sending sensitive information like passwords, forms, etc.

3. Difference Between doGet() and doPost() (Minimum 4 Differences)

doGet()	doPost()
Sends data through URL	Sends data in request body
Data is visible in URL	Data is hidden from URL
Limited amount of data can be sent	Large amount of data can be sent
Less secure	More secure
Request can be bookmarked/cached	Cannot be bookmarked/cached
Mainly for retrieving data	Mainly for submitting data
Faster (no body parsing)	Slightly slower (body processing)

Q1 (b) What is XML DTD? Explain with example. Differentiate XML DTD vs XML Schema (Min 4). [9 Marks]

Ans:

DTD = Document Type Definition.

It defines **rules, structure, elements, attributes** of an XML document.

1. What is XML DTD? (Explanation)

1. Defines Structure of XML Document

It specifies the allowed elements, attributes, and their order.

2. Validates XML Document

Ensures XML follows correct structure → known as **valid XML**.

3. Works Like a Grammar

Defines what tags are allowed and how many times they can appear.

4. Two Types of DTD

- **Internal DTD** (inside XML file)
- **External DTD** (placed in a separate .dtd file)

5. Ensures Consistency of XML Data

Helps maintain proper structure across multiple XML files.

2. XML DTD Example

Internal DTD Example:

```
<!DOCTYPE student [  
  <!ELEMENT student (name, age)>  
  <!ELEMENT name (#PCDATA)>  
  <!ELEMENT age (#PCDATA)>  

```

```
<student>  
  <name>Vijay</name>  
  <age>21</age>  
</student>
```

student element must contain name and age

3. Difference Between XML DTD and XML Schema (Minimum 4 Differences)

XML DTD	XML Schema (XSD)
Written in non-XML format	Written in XML syntax
Cannot define data types properly	Supports rich data types (int, float, date)
Less powerful and less strict	More powerful and strict validation
No namespace support	Supports XML namespaces
Hard to read and extend	Easy to read, modify, and extend
Limited structure rules	Provides advanced rules (minOccurs, maxOccurs)

Both name and age contain only text (#PCDATA)

Q2 (a) Explain the servlet lifecycle. Explain session management using cookies and URL rewriting. [12 Marks]

Ans:

Servlet lifecycle means the complete process of **creating, using, and destroying a servlet**.

Session management keeps track of **individual users** across multiple requests.

Techniques: **Cookies & URL Rewriting**.

1. Servlet Lifecycle

Servlet lifecycle has **5 major steps**:

1) Loading the Servlet Class

- Servlet container (Tomcat) loads the servlet class into memory.
- Happens when the first request comes or during server startup (if configured).

2) Creating the Servlet Instance

- Container creates **only one object** of the servlet.
- This object is shared for all requests → improves performance.

3) Initialization using `init()`

- Container calls the `init()` method **only once**.
- Used for initializing resources like:
 - Database connection
 - File reading
 - Loading configuration parameters

```
public void init() {  
    // initialization code  
}
```

4) Request Handling using service()

- Called **for every client request**.
- Determines whether request is **GET** or **POST**:
 - Calls doGet() for GET
 - Calls doPost() for POST

```
public void service() {  
    // request handling code  
}
```

5) Destroying the Servlet using destroy()

- Called once before container unloads servlet.
- Used for cleanup activities:
 - Closing DB connections
 - Releasing memory resources

```
public void destroy() {  
    // cleanup code  
}
```

2. Session Management Using Cookies

Concept:

Cookies store small pieces of data in the user's browser to identify user sessions.

How Cookies Manage Session (Points):

1. Server Creates Cookie Object

```
Cookie c = new Cookie("userid", "101");
```

2. Cookie Sent to Browser

```
response.addCookie(c);
```

3. Browser Stores the Cookie

- Stored as a text file.

4. Browser Returns Cookie Automatically

- For every next request, cookie is sent back.

5. Server Identifies User

- Session maintained based on cookie values.

3. Session Management Using URL Rewriting

Concept:

URL rewriting attaches session information **inside the URL** when cookies are disabled.

How URL Rewriting Works (Points):

1. Session ID or data added in URL

Example:

`http://mysite.com/home?session=1234`

2. Client clicks the link with session data

3. Browser sends session ID to server

4. Server extracts the session ID

5. User session is tracked without cookies

Example Code:

```
String url = response.encodeURL("home.jsp");  
out.println("<a href='" + url + "'>Go to Home</a>");
```

Q2 (b) Write a note on AJAX. [6 Marks]

Ans:

AJAX = **Asynchronous JavaScript and XML**.

It updates parts of a webpage **without reloading the entire page**.

1. Meaning

- AJAX is not a language; it's a technique combining:
 - JavaScript
 - XML/JSON
 - HTML
 - CSS
 - DOM

2. Asynchronous Communication

- Requests are sent to the server **in the background**.
- Webpage remains visible and active.

3. No Page Refresh

- Only small parts of the page update.
- Improves user experience.

4. XMLHttpRequest Object

- Key component used to send and receive data.
- Modern websites use **Fetch API** too.

5. Data Formats Supported

- JSON (most common)
- XML
- HTML
- Plain text

6. Working of AJAX (Steps)

1. User triggers an event (button click).
2. JavaScript creates XMLHttpRequest object.
3. Request is sent to server asynchronously.
4. Server processes the request.
5. Server sends data back in JSON/XML.
6. JavaScript updates specific part of webpage.

7. Real-Life Examples

- Google search suggestions
- Auto-refresh notifications
- Live chat apps
- Online forms (auto validation)

Q3 a) Explain the JSP support for MVC paradigm.[8]

Ans:

MVC = Model – View – Controller

JSP supports MVC by acting mainly as the **View layer**, while Servlets act as **Controllers** and JavaBeans/Classes act as **Models**.

This separation increases clarity, scalability and maintainability of web applications.

JSP Support for MVC – Detailed Explanation

1 JSP as the View

- JSP is used to **display output** to the user.
- Contains **HTML, CSS, JSTL, Expression Language**.
- JSP should not contain business logic; it only receives data from the controller and shows it.

2 Servlet as the Controller

- Servlet receives the client request.
- Performs:
 - Input validation
 - Calling business logic (Model)
 - Preparing data for JSP
- Finally, it forwards the request to JSP using:

```
RequestDispatcher rd = request.getRequestDispatcher("result.jsp");  
rd.forward(request, response);
```

3 JavaBeans / Model Classes

- Represents application data / business logic.
- Handles:
 - Database operations
 - Calculations
 - Business rules

Example Bean:

```
public class Student {  
    int id;  
    String name;  
    // getters & setters  
}
```

4 JSP retrieves Model data using Expression Language / JSTL

Example:

```
Student s = (Student) request.getAttribute("std");  
<p>Name: ${std.name}</p>
```

5 Clean separation of responsibilities

- Controller = Request processing
- Model = Business logic
- JSP = View

Q3 b) Explain Struts framework with respect to architecture, actions, interceptors & exception handling.[9]

Ans:

1. Struts Architecture

Struts is a **Java MVC framework**.

Draw this simple diagram in exam:

Browser



ActionServlet (Controller)



struts.xml (Routing)



Interceptors



Action Class (Model/Logic)



Result (JSP View)

Components Explained

1 ActionServlet (Controller)

- Receives all incoming requests.
- Reads mappings from struts.xml.
- Calls interceptor stack & Action classes.

2 struts.xml

- Main configuration file.
- Maps URL → Action class → JSP view.

Example:

```
<action name="login" class="com.demo.LoginAction">
  <result name="success">home.jsp</result>
  <result name="error">login.jsp</result>
</action>
```

3 Action Classes

- Contain business logic.
- Must return a result string (e.g., “success”, “error”).

Example:

```
public class LoginAction extends ActionSupport {  
    public String execute() {  
        return "success";  
    }  
}
```

4 Interceptors

- Work like filters.
- Execute **before** and **after** Action class.
- Predefined interceptors:
 - Validation interceptor
 - Logging interceptor
 - Exception interceptor
 - Parameter interceptor

Purpose: Authentication, logging, pre/post processing.

5 Exception Handling

Handled using:

- Exception interceptors
- Exception mappings in struts.xml

Example:

```
<global-exception-mapping exception="java.lang.Exception"  
result="errorPage"/>
```

This redirects user to errorPage.jsp on exception.

OR

Q4 a) Explain JSP lifecycle. Differentiate JSP vs Servlet (min. 4).[9]

Ans: 1. JSP Lifecycle

1 Translation

- JSP is converted to a servlet by the container.

2 Compilation

- The generated servlet is compiled into .class.

3 Class Loading

- The .class file is loaded into memory.

4 Object Instantiation

- Container creates servlet object for JSP.

5 Initialization – `jspInit()`

- Called only once.

6 Request Processing – `_jspService()`

- Called for every client request.
- Performs main processing of JSP.

7 Destruction – `jspDestroy()`

- Called when server shuts down.

2. JSP vs Servlet (Minimum 4 Differences)

JSP	Servlet
Mostly contains HTML with little Java.	Pure Java code.
Used for View layer in MVC.	Used for Controller layer.
Easier to write UI pages.	Harder to write HTML using <code>println()</code> .
Has many implicit objects (out, request, session).	No implicit objects; must create manually.
Translated into Servlet automatically.	Developer writes servlet directly.
Better for rapid UI development.	Better for processing logic.

Q4 b) Explain the concept of Web Services. Explain WSDL & SOAP.[8]

Ans:

Web services allow applications to communicate over the internet using **standard protocols** like HTTP, XML, SOAP, REST.

Characteristics

- Platform-independent
- Language-independent
- Machine-to-machine communication
- Used in cloud, mobile apps, distributed systems

2. WSDL (Web Services Description Language)

- It is an **XML-based file** that describes everything about a web service.
- Defines:
 - Service name
 - Methods / operations
 - Input-output parameters
 - Protocol (SOAP/HTTP)
 - URL of the service

Example (Simple)

```
<wsdl:operation name="getStudent">
  <input message="studentRequest"/>
  <output message="studentResponse"/>
</wsdl:operation>
```

3. SOAP (Simple Object Access Protocol)

- An **XML-based messaging protocol** used to send data between client and server.
- Works over HTTP, SMTP.
- Very secure and reliable.

SOAP Message Structure

```
<Envelope>
  <Header>...</Header>
  <Body>
    <getInfo>
      <id>101</id>
    </getInfo>
  </Body>
</Envelope>
```

Q5 a) Explain various types of Arrays in PHP. Explain each with example code. (9 marks)

Ans:

An array in PHP is a special variable that can hold multiple values under a single name. PHP supports mainly **three types of arrays**.

1) Indexed Array (Numeric Array)

Definition

- Stores values with numeric indexes (0,1,2...)
- Used when we only need a list of values.

Example

```
$fruits = array("Apple", "Banana", "Mango");
```

```
echo $fruits[0]; // Output: Apple
```

Looping

```
foreach($fruits as $f){  
    echo $f . "<br>";  
}
```

2) Associative Array

Definition

- Uses **named keys** instead of numbers.
- Good for storing data in key-value format.

Example

```
$student = array(  
    "name" => "Rohan",  
    "age" => 21,  
    "dept" => "CSE"  
);
```

```
echo $student["name"]; // Output: Rohan
```

3) Multidimensional Array

Definition

- Array containing one or more arrays.
- Used for table-like structures.

Example

```
$employees = array(  
    array("101", "Amit", "HR"),  
    array("102", "Riya", "IT"),  
    array("103", "Kunal", "Sales")  
);
```

```
echo $employees[1][1]; // Output: Riya
```

Use

- Best for storing multiple records (similar to database rows).

Q5 b) i) WAP & WML (4 marks)

Ans:

WAP (Wireless Application Protocol)

Definition

WAP is a protocol used earlier to allow mobile phones to access the internet using small screens, low bandwidth, and simple browsers.

Features

- Uses WML pages
- Uses a micro-browser
- Uses WAP gateway for conversion
- Lightweight and optimized for mobile devices

WML (Wireless Markup Language)

Definition

WML is a markup language (like HTML) used to design WAP pages.

Features

- Uses deck and card structure
- Displays text, links, simple input
- Suitable for small-screen devices

Example

```
<?xml version="1.0"?>
<wml>
  <card id="home" title="Welcome">
    <p>Hello WAP User!</p>
  </card>
</wml>
```

Q5 b) ii) C# Vs Java (5 marks)

Ans:

Both are object-oriented languages but differ in platform, memory handling, and features.

Difference Table

C#	JAVA
Developed by Microsoft	Developed by Sun/Oracle
Runs on .NET Framework	Runs on JVM (Java Virtual Machine)
Supports multiple programming styles	Purely Object-Oriented
Uses properties, events, delegates	Uses interfaces, method overloading
Specialized for Windows Apps + Web (ASP.NET)	Cross-platform, mobile apps (Android)
Memory management with automatic garbage collection	Garbage collection + JVM tuning
Strong integration with Visual Studio	Strong integration with Eclipse/IntelliJ

OR -Q6 a) Explain PHP with MySQL using example. (6 marks)

Ans:

PHP is used to write server-side logic, and MySQL stores data. Together they create dynamic web applications.

Steps to Connect PHP with MySQL

1) Create Database

Example: employee

2) Connect to MySQL

```
$conn = mysqli_connect("localhost", "root", "", "employee");
```

3) Run SQL Query

```
$result = mysqli_query($conn, "SELECT * FROM emp");
```

4) Display Output

```
while($row = mysqli_fetch_assoc($result)){  
    echo $row['emp_name']." - ".$row['emp_dept']."<br>";  
}
```

5) Close Connection

```
mysqli_close($conn);
```

This shows how PHP fetches and displays database records.

Q6 b) Write note on: (12 marks)

Ans:

i) Session Tracking in PHP

Definition

Sessions store user-specific data on the server across multiple pages.

Why needed?

- HTTP is stateless
- Sessions maintain login info, cart items, preferences

Start a Session

```
session_start();
```

Store Data

```
$_SESSION['user'] = "Vijay";
```

Access Data

```
echo $_SESSION['user'];
```

Destroy Session

```
session_destroy();
```

ii) .NET Framework

Concept

.NET Framework is a software platform by Microsoft for building applications (web, desktop, mobile).

Features

- Supports languages (C#, VB.NET, F#)
- CLR handles memory, security, execution
- Large class library
- ASP.NET for web apps
- ADO.NET for database access

Benefits

- Fast development
- Secure
- Scalable apps
- Easy debugging

iii) Node.js

Definition

Node.js is a server-side runtime that runs JavaScript outside the browser.

Features

- Non-blocking I/O (handles thousands of users)
- Event-driven architecture
- Built on V8 engine
- Uses **npm** (largest package manager)
- Ideal for chat apps, streaming, APIs

Uses

- Real-time apps
- API servers
- Backend of web apps
- IoT applications

Q7 (a) Explain Ruby with its advantages. Explain control statements in Ruby. [10]

Ans:

1. What is Ruby?

Ruby is a **pure object-oriented, dynamic, interpreted programming language** developed by *Yukihiro Matsumoto*.

It focuses on **simplicity, readability, and developer happiness**.

Ruby is widely used for writing scripts and for web development using **Ruby on Rails**.

2. Advantages of Ruby

(i) Pure Object-Oriented

Everything in Ruby is an object—even numbers, strings, and classes.

(ii) Easy and Readable Syntax

Ruby code looks like English, making it very easy to learn.

(iii) Dynamic and Flexible

You can add methods, modify classes, and change behavior at runtime.

(iv) Automatic Memory Management

Garbage collection is built-in, reducing programmer effort.

(v) Large Standard Library

Ruby provides many built-in modules for networking, file handling, text processing, etc.

(vi) Supports Mixins

Multiple inheritance is implemented through modules, making code reusable.

(vii) Strong Web Support

Ruby on Rails is one of the most productive web frameworks.

3. Control Statements in Ruby

Control statements decide the flow of execution.

(i) if–elsif–else

```
age = 20
```

```
if age < 18
  puts "Minor"
elsif age >= 18 && age < 60
  puts "Adult"
else
  puts "Senior"
end
```

(ii) unless

Opposite of if → executes when condition is false.

```
temp = 30
puts "It is not cold" unless temp < 10
```

(iii) case–when

Used for multiple choice selection.

```
day = 3

case day
when 1 then puts "Monday"
when 2 then puts "Tuesday"
when 3 then puts "Wednesday"
else puts "Invalid"
end
```

(iv) Loops

while loop

```
i = 1
while i <= 5
  puts i
  i += 1
end
```

for loop

```
for x in 1..3  
  puts x  
end
```

times loop

```
5.times { puts "Hello" }
```

Q7 (b) Explain EJB concept & five basic examples of using EJB. [7]

Ans:

1. What is EJB?

EJB (Enterprise Java Bean) is a **server-side component** used to build **large-scale, secure, distributed, transactional enterprise applications**.

It runs inside an **EJB container** that provides:

- Security
- Transaction management
- Persistence support
- Concurrency handling
- Life-cycle management

2. Types of EJB

1. **Session Beans (Stateless & Stateful)**
2. **Entity Beans**
3. **Message-Driven Beans (MDB)**

3. Five Basic Examples Where EJB is Used

Example 1: Online Banking System

EJB manages:

- Money transfer
- Account validation
- Secure transactions

Example 2: E-Commerce / Shopping Cart

Used for:

- Product catalog
- Order processing
- Payment handling

Example 3: Reservation Systems (Airline / Hotel)

EJB handles:

- Ticket booking
- Availability checking
- Seat allocation

Example 4: Hospital Management System

EJB manages:

- Patient records
- Billing
- Appointment scheduling

Example 5: Inventory Management System

EJB handles:

- Stock monitoring
- Supplier updates
- Purchase orders

OR-Q8 (a) Explain arrays in Ruby. Explain Rails with AJAX. [10]

Ans:

1. Arrays in Ruby

Ruby arrays store **ordered, indexed collection of objects**, and can store mixed data types.

Creating arrays

```
arr = [10, "hello", 3.14]
```

Accessing elements

```
puts arr[0] # 10  
puts arr[1] # hello
```

Adding elements

```
arr << "new"
```

Removing elements

```
arr.delete("hello")
```

Iterating

```
arr.each do |item|  
  puts item  
end
```

2. Rails with AJAX

AJAX = Asynchronous JavaScript and XML → updates web page **without full reload**.

How Rails supports AJAX

1. Use remote: true in forms/links
2. Rails sends asynchronous request
3. Controller responds with .js.erb file
4. JS updates web page dynamically

Example

Form (AJAX request)

```
<%= form_with url: "/students", remote: true do |f| %>
  <%= f.text_field :name %>
  <%= f.submit "Add" %>
<% end %>
```

Controller

```
def create
  @student = Student.create(params[:student])
  respond_to do |format|
    format.js
  end
end
```

create.js.erb

```
alert("Student added successfully!");
```

Q8 (b) Explain Document Request in Rails. [4]

Ans:

Document Request = how Rails processes incoming browser requests.

Steps in Document Request Processing

1. Browser sends URL Request

User enters:

/students/show/5

2. Rails Router (routes.rb) matches the URL

Router decides the controller and action.

Example:

get 'students/show/:id', to: 'students#show'

3. Controller Action Executes

Controller loads:

- model data
- processes logic

4. View is Rendered

Rails automatically renders:

views/students/show.html.erb

5. Response sent back to browser

HTML page is displayed.

Q8 (c) Explain advantages of Ruby and Rails. [3]

Ans:

Advantages of Ruby

1. Simple, readable English-like syntax
2. Pure object-oriented
3. Built-in garbage collection

Advantages of Rails

1. Rapid Development (Convention over Configuration)
2. Built-in support for MVC
3. Large community and many ready-made libraries (gems)

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